What’s in a Name?

Time, place and sex in American first names, 1880–2025

Project proposal · Data Science Methods for Management · MSc Actuarial Science, University of Lausanne
Group members: [to be completed] · 19 October 2026

1. Background and Motivation

We chose this topic because our group shares an interest in demography, and first names offer a simple, everyday way to see how a population changes over time. Every child receives a name at birth, so names reflect the tastes and habits of each generation, each region and each sex. In the United States, the Social Security Administration has published the number of babies given each name every year since 1880, which gives more than a century of data covering almost every birth. The topic interests us because it raises questions anyone can relate to: which names are passing fashions and which last, whether states still have their own naming habits, and whether names given to both boys and girls stay that way. It also matters in practice, because names are often used to guess a person’s sex or background in surveys, research and customer databases, and those guesses are only as reliable as the link between a name and the people who carry it. From a data science point of view, the project fits the course well: the data need cleaning and reshaping, names can be treated as text, and the questions can be answered with visualisation, regression and classification models, and clustering.

2. Project Objectives

The overall objective is to describe and model how American first names vary with the time, place and sex of birth, and how these differences have changed since 1880, combining exploratory analysis and visualisation, predictive modelling and unsupervised learning. For time, we aim to identify fashion cycles by clustering popularity trajectories to separate short-lived fads from enduring names, and to test whether the letters of a name predict the period in which it was popular. For place, we aim to identify the names that characterise each state using TF-IDF, map naming diversity over time and measure whether state profiles have converged towards a national pattern. For sex, we aim to track the female share of each name, cluster these trajectories, assess whether names that change sex do so asymmetrically or largely remain stable, and test with a classification model whether letter patterns help anticipate such changes. Finally, we aim to communicate the findings to a general audience through a reproducible Quarto report and a short video presentation.

3. Research Questions

Our main question is how American first names vary with the time, place and sex of birth, and how these differences have changed since 1880. RQ1 (time) asks whether names follow identifiable life cycles, from short-lived fads to timeless classics, and whether the spelling of a name carries information about the period in which it was popular; it will be addressed by clustering normalised popularity trajectories from the national data and by a regression predicting each name's mean year of birth from its character n-grams, evaluated on names not seen during training. RQ2 (place) asks whether each state has a distinctive naming profile and whether state profiles have converged towards a national pattern since 1910; each state-decade will be treated as a document so that TF-IDF identifies characteristic names, alongside diversity indicators and each state's distance from the national distribution. RQ3 (sex) asks whether names move between sexes over time, whether this movement is asymmetric, as Lieberson, Dumais and Baumann (2000) argued, or largely stable, as Seguin, Julien and Zhang (2021) found, and whether it can be anticipated from the letters of a name; it will be addressed by clustering female-share trajectories and by a classification model trained up to a cut-off year to predict later shifts. Because the data are aggregate counts with no information on parents, the project describes and predicts patterns but makes no causal claims about why parents choose particular names.

4. Data Source

All data are public, free and available without registration, and will be downloaded directly and read with pandas, with no scraping or API access required. The main source is the SSA baby names data (https://www.ssa.gov/oact/babynames/limits.html), released under a CC0 licence, which includes a national file with births by first name and sex from 1880 to 2025 and a state-level file with births by first name, sex and state from 1910 to 2025. A public table from the US Census Bureau assigning each state to its region and division will be merged by state. The main challenges are that names with fewer than five occurrences in a year (or in a state and year) are suppressed for privacy, which understates diversity, especially in small states, so we will test sensitivity to minimum-count thresholds; that Social Security card applications began in the late 1930s, so early birth cohorts are represented only by people who applied later in life; that sex is recorded as a binary category; and that spelling variants such as Caitlin and Kaitlyn count as distinct names. The data contain no individual or parental information, so conclusions concern population-level patterns only, and no inference will be made about the identity or origin of individuals.

5. Data Description

The national data consist of one comma-separated file per year with three fields (name, sex and number of births), about two million rows in total, and the state data consist of one file per state with five fields (state, sex, year, name and number of births), several million rows in total. The original variables are the first name (text), sex and state (categorical), year of birth (date) and number of births (numeric). Derived variables will include each name's share of births within its year and sex, its popularity rank and its female share, name length and character n-grams, trajectory features such as peak year, years in use and mean year of birth, diversity indicators by state and decade, TF-IDF scores, distance to the national profile and cluster membership, while Census region and division will be merged in, so that the analysis dataset covers all required variable types. Preprocessing will involve stacking the yearly and state files with the year as a column, converting counts into shares so that trends are not driven by changes in total births, combining male and female rows to compute female shares, setting minimum-count thresholds for trajectory analyses, aggregating state data by decade to build state-decade by name matrices for TF-IDF, and vectorising names into character n-grams, with training and test sets split by name and temporal hold-outs for the classification task. The analysis will be carried out in Python (pandas, scikit-learn, matplotlib and seaborn) and documented in Quarto.

6. Schedule and Timeline

The schedule follows the order of the course syllabus, so that each block is applied to the project shortly after it is taught. After submitting the proposal on 19 October, we will download the data, audit its structure and the effect of suppression, and build the cleaning pipeline and derived variables between 20 October and 2 November. From 3 to 15 November we will carry out the exploratory analysis and visualisation of fashion cycles, state profiles and female-share trajectories, and on 16 November we will submit the project update in Quarto HTML format. From 17 November to 5 December we will cluster the trajectories, compute TF-IDF and convergence measures, fit and validate the regression and classification models, and run robustness checks. From 6 to 14 December we will interpret the results, finalise the visualisations and write the discussion and limitations, and from 15 to 20 December we will review the report, prepare the slides and record the video, before submitting the final Quarto/HTML report and video on 21 December. One member will lead data preparation and the integration of the report, each of the other three will lead one dimension (time, place or sex), and methods and conclusions will be reviewed jointly.

References

Lieberson, S., Dumais, S., & Baumann, S. (2000). The instability of androgynous names: The symbolic maintenance of gender boundaries. *American Journal of Sociology*, 105(5), 1249–1287.

Seguin, C., Julien, C., & Zhang, Y. (2021). The stability of androgynous names: Dynamics of gendered naming practices in the United States 1880–2016. *Poetics*, 85, 101501. https://doi.org/10.1016/j.poetic.2020.101501

Social Security Administration. Beyond the Top 1000 Names: national and state-specific data. https://www.ssa.gov/oact/babynames/limits.html